

Nurture Course

PHASE : SPEED & ACCURACY TEST

TARGET : PRE - MEDICAL 2023

Test Type : **PRACTICE TEST**

Test Pattern : **NEET (UG)**

TEST DATE :

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	3	4	4	2	1	3	1	4	3	3	3	3	1	1	1	1	3	2	3	3	3	2	1	2	4	2	4	1	2	4
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45															
A.	2	4	3	1	2	2	1	4	1	3	2	3	2	2	2															

HINT - SHEET



Click Here To Join our Telegram Channel

1. **Ans (3)**

$$\omega = 60^\circ/\text{s} = \frac{\pi}{3} \text{ rad/s}$$

$$v = r\omega = (3.5) \frac{\pi}{3} \text{ m/s} = \frac{7\pi}{6} \text{ m/s}$$

2. **Ans (4)**

$$\omega_i = 12 \text{ rad/s},$$

$$\omega_F = \left(\frac{540}{\pi} \times \frac{27}{60} \right) \text{ rad/s}$$

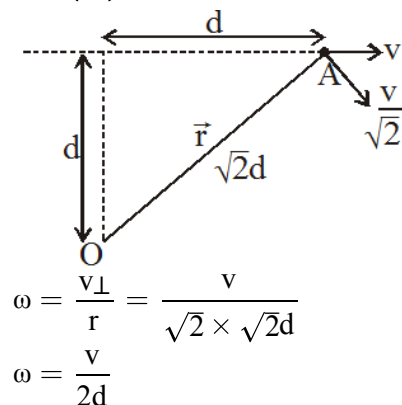
$$= 18 \text{ rad/s}$$

$$\omega_F = \omega_i + \alpha t$$

$$\alpha = \frac{\omega_F - \omega_i}{t}$$

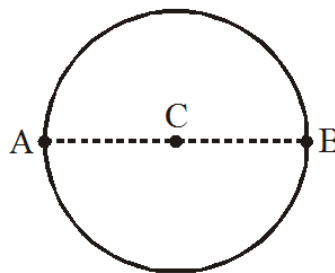
$$= \frac{18 - 12}{2} = 3 \text{ rad/s}^2$$

3. **Ans (4)**



4. **Ans (2)**

$$\frac{\omega_A}{\omega_C} = \frac{\frac{v_A}{r_A}}{\frac{v_C}{r_C}} = \frac{r_C}{r_A}$$



$$= \frac{a}{2a} = \frac{1}{2}$$

5. **Ans (1)**

Velocity is tangential to path.

6. **Ans (3)**

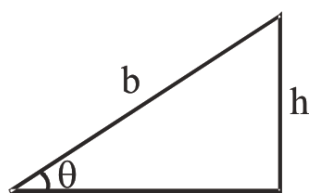
Angular velocity of the second hand

$$\omega = \frac{2\pi}{60} \text{ rad/s}, v = r\omega = 3 \times \frac{2\pi}{60} = \frac{\pi}{10} \text{ cm/s}$$

$$\Delta \vec{v} = \Delta \vec{v}_2 - \Delta \vec{v}_1 = 2v \sin \left(\frac{\theta}{2} \right)$$

$$\Delta v = \sqrt{2}v = \sqrt{2} \times \frac{\pi}{10} = \frac{\pi}{5\sqrt{2}} \text{ cm/s} \quad (\because \theta = 90^\circ)$$

7. **Ans (1)**

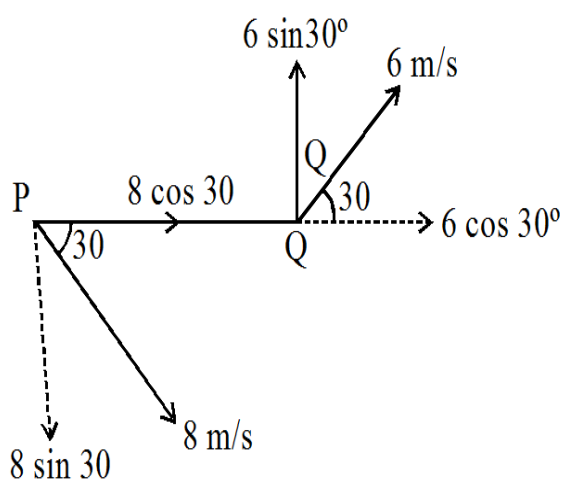


$$\tan \theta \simeq \sin \theta = \frac{h}{b}$$

$$\therefore \frac{h}{b} = \frac{v^2}{Rg}$$

$$\therefore h = \frac{v^2 b}{Rg}$$

8. **Ans (4)**



Relative velocity of P w.r.t. Q

perpendicular to line PQ

$$V_{PQ} = 8 \sin 30 - (-6 \sin 30) = 4 + 3 = 7 \text{ m/s}$$

$$\omega_{PQ} = \frac{V_{PQ}}{PQ} = \frac{7}{10} = 0.7 \text{ rad/sec}$$

9. **Ans (3)**

$$(V_{\text{avg}})_A = \frac{R}{2} \text{ m/s} ; (V_{\text{avg}})_B = \frac{R}{5} \text{ m/s}$$

$$\frac{(V_{\text{avg}})_A}{(V_{\text{avg}})_B} = \frac{5}{2} \text{ or } 5 : 2$$

10. **Ans (3)**

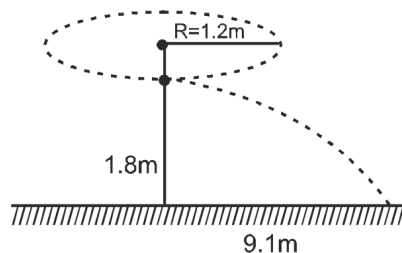
$$R = 30 \text{ cm}, v = 2t$$

$$a_R = \frac{v^2}{R} \text{ and } a_T = \frac{dv}{dt}$$

$$a_R = \frac{4t^2}{R} \text{ and } a_T = 2 \text{ m/s}^2$$

$$\text{at } t = 3 \text{ sec, } a_R = \frac{4(3)^2}{30 \times 10^{-2}} = 120 \text{ m/s}^2$$

11. **Ans (3)**



The time taken to fall on ground

$$= \sqrt{\frac{2 \times 1.8}{9.8}} = \sqrt{\frac{36}{98}}$$

velocity at time of string breaks

$$v = \frac{\text{dis tan ce}}{\text{time}} = 9.1 \sqrt{\frac{98}{36}}$$

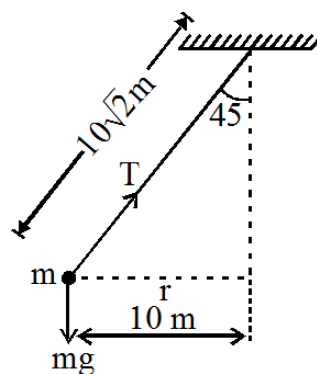
$$\text{Centripetal acceleration} = \frac{v^2}{R}$$

$$= \frac{9.1 \times 9.1 \times 98}{1.2 \times 36} = 187.856 = 188 \text{ m/s}^2$$

12. **Ans (3)**

$$T_{OA} = 2m\omega^2 r + T_{AB} \text{ and } T_{AB} = 2rm\omega^2.$$

14. **Ans (1)**



$$\tan \theta = \frac{v^2}{rg} \Rightarrow \tan 45^\circ = \frac{v^2}{10 \times 10} \Rightarrow v = 10$$

$$T = \frac{2\pi r}{v} = \frac{2\pi \times 10}{10} = 2\pi \text{ sec}$$

16. **Ans (1)**

For equilibrium

$$\mu \geq \tan \alpha \Rightarrow \frac{1}{3} \geq \tan \alpha \Rightarrow \frac{1}{3} \geq \frac{1}{\cot \alpha} \Rightarrow \cot$$

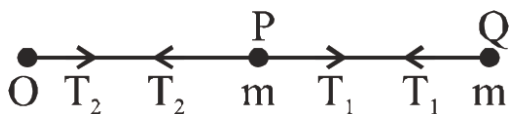
$$\alpha \leq 3$$

17. **Ans (3)**

$$a = \sqrt{a_c^2 + a_t^2} = \sqrt{\left(\frac{V^2}{r}\right)^2 + a_t^2}$$

$$a = \sqrt{\left(\frac{900}{450}\right)^2 + (2)^2} = 2.8 \text{ m/s}^2$$

18. **Ans (2)**



$$T_1 = m\omega^2(2) \quad \dots\dots(1)$$

$$T_2 - T_1 = m\omega^2(1)$$

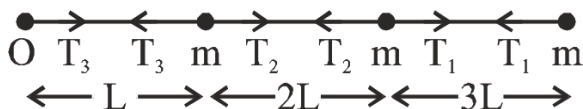
$$T_2 = T_1 + m\omega^2$$

$$T_2 = m\omega^2(2) + m\omega^2$$

$$T_2 = 3m\omega^2 \quad \dots\dots(2)$$

$$\frac{T_{PQ}}{T_{PO}} = \frac{T_1}{T_2} = \frac{2}{3}$$

19. **Ans (3)**



$$T_1 = m\omega^2(6L) \quad \dots\dots(1)$$

$$T_2 - T_1 = m\omega^2(3L)$$

$$T_2 = T_1 + m\omega^2(3L)$$

$$T_2 = m\omega^2(6L) + m\omega^2(3L)$$

$$T_2 = m\omega^2(9L) \quad \dots\dots(2)$$

$$T_3 - T_2 = m\omega^2(L)$$

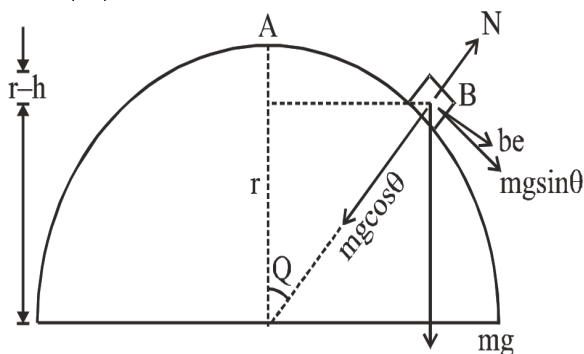
$$T_3 = T_2 + m\omega^2(L)$$

$$T_3 = m\omega^2(9L) + m\omega^2L$$

$$T_3 = m\omega^2(10L) \quad \dots\dots(3)$$

$$T_1 : T_2 : T_3 :: 6 : 9 : 10$$

20. **Ans (3)**



By newton law

By conservation

of mechanical energy between A & B

$$Mg \cos \theta - N = \frac{mV^2}{r}$$

$$\frac{1}{2}mV^2 = mg(r-h)$$

$$mg \cos \theta = \frac{mV^2}{r}$$

$$\frac{V^2}{g} = 2(r-h) \quad \dots(2)$$

$$\cos \theta = \frac{V^2}{rg}$$

$$\frac{h}{r} = \frac{V^2}{rg}$$

By (1) & (2)

$$h = 2(r-h)$$

$$\Rightarrow h = 2r - 2h$$

$$\Rightarrow 3h = 2r \Rightarrow h = \frac{2r}{3}$$

21. **Ans (3)**

$$T = m\omega^2 r$$

$$2mg = m\omega^2 r$$

$$\omega = \sqrt{\frac{2g}{r}} = \sqrt{2g}$$

22. **Ans (2)**

Let 'l' be the original length.

$$\text{Given } m\omega^2(l+x_1) = kx_1$$

$$m(2\omega)^2(l+x_2) = kx_2$$

$$\text{where } x_1 = 1 \text{ cm } x_2 = 5 \text{ cm}$$

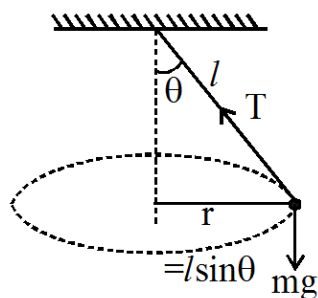
$$\therefore \frac{4(l+x_2)}{l+x_1} = \frac{x_2}{x_1}$$

$$4x_1(l+x_2) = x_2(l+x_1)$$

$$4(\ell+5) = 5(\ell+1)$$

$$\ell = 15 \text{ cm}$$

23. Ans (1)



$$T \sin \theta = m \omega^2 r$$

$$T \sin \theta = m \omega^2 l \sin \theta$$

$$T = m \omega^2 l$$

24. Ans (2)

$$k = as^2$$

$$\frac{1}{2} m v^2 = as^2$$

$$v^2 = \frac{2as^2}{m} \quad \dots(1)$$

Centripetal acceleration Tangential
acceleration

$$a_c = \frac{v^2}{R} = \frac{2as^2}{mR}$$

diff. eqⁿ (1) w.r.t. t

$$2v \cdot \frac{dv}{dt} = \frac{2a}{m} (2S) \frac{ds}{dt}$$

$$v \frac{dv}{dt} = \frac{2a}{m} S v$$

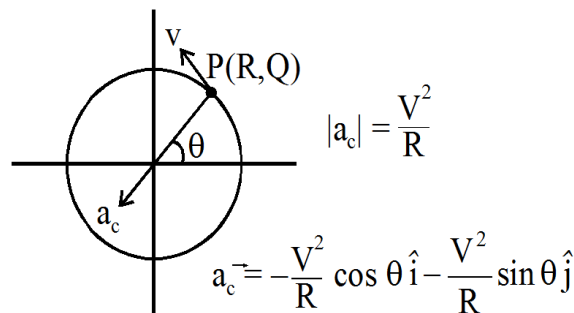
$$a_T = \frac{dv}{dt} = \frac{2a}{m} S$$

$$a = \sqrt{a_c^2 + a_T^2}$$

$$a = \sqrt{\left(\frac{2as^2}{mR}\right)^2 + \left(\frac{2aS}{m}\right)^2} = 2aS \sqrt{\frac{S^2}{R^2} + 1}$$

$$\Rightarrow a = 2aS \left(1 + \frac{S^2}{R^2}\right)^{1/2}$$

25. Ans (4)



26. Ans (2)

$$\tan \theta = \frac{v^2}{rg}, \quad v = 60 \times \frac{5}{18} = \frac{50}{3} \text{ m/s}$$

$$r = 0.1 \text{ km} = 100 \text{ m}$$

$$\theta = \tan^{-1} \left(\frac{(50/3)^2}{100 \times 9.8} \right)$$

27. Ans (4)

$$\tan \theta = \frac{V^2}{rg}$$

$$V = \sqrt{rg \tan \theta}$$

$$= \sqrt{90 \times 10 \times \tan 45^\circ}$$

$$= 30 \text{ m/s}$$

28. Ans (1)

$$f = \mu N \geq \frac{mV^2}{R} \Rightarrow$$

$$\mu mg \geq \frac{mV^2}{R} \Rightarrow V \leq \sqrt{\mu Rg}$$

$$\text{for maximum } V = \sqrt{\mu Rg}$$

$$= \sqrt{0.8 \times 84.5 \times 10} = 26 \text{ m/s}$$

29. Ans (2)

$$V \leq \sqrt{\mu Rg}$$

$$V = \sqrt{0.6 \times 150 \times 10}$$

$$V = 30 \text{ m/s}$$

30. Ans (4)

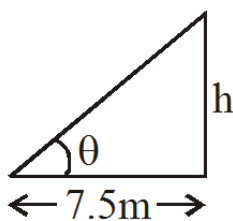
$$\tan \theta = \frac{V^2}{rg}$$

$$= \frac{(14\sqrt{3})^2}{20\sqrt{3} \times 9.8} = \frac{196 \times 3}{20\sqrt{3} \times 9.8} = \sqrt{3}$$

$$\Rightarrow \theta = 60^\circ$$

31. **Ans (2)**

$$\tan \theta = \frac{V^2}{rg}$$



$$\frac{h}{7.5} = \frac{(10)^2}{(150)(10)}$$

$$\frac{h}{7.5} = \frac{100}{1500} \Rightarrow h = \frac{7.5}{15} \Rightarrow h = 0.50 \text{ m}$$

32. **Ans (4)**

for a complete loop velocity at B should be

greater than equal to $= \sqrt{5gR}$

$$V_B \geq \sqrt{5gR} \Rightarrow \sqrt{2gh} \geq \sqrt{5gR}$$

$$R \leq \frac{2h}{5}$$

$$R_{\max} = \frac{2h}{5} = 2\text{cm}$$

34. **Ans (1)**

$$T_{\max} = T_L = m\omega^2 r + mg$$

$$30 = 0.5\omega^2(2) + 0.5 \times 10$$

$$25 = \omega^2$$

$$\omega = 5 \text{ rad/sec}$$

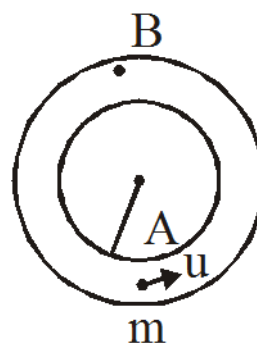
35. **Ans (2)**

$$V_{\min} \text{ at P} = \sqrt{Lg}$$

$$X = u \sqrt{\frac{2H}{g}} \Rightarrow X = \sqrt{Lg} \sqrt{\frac{2(2L)}{g}}$$

$$X = 2L$$

36. **Ans (2)**



By conservation of mechanical energy

$$\frac{1}{2} mV^2 = mg(2r)$$

$$V = \sqrt{4gr}$$

37. **Ans (1)**

For position B

$$mg - N = \frac{mV_B^2}{R}$$

for value of R

$$\text{When } N = 0 \quad mg \geq \frac{mV_B^2}{R}$$

$$R \geq \frac{V_B^2}{g} \text{ and } mg(h_A - h_B) = \frac{1}{2} mV_B^2$$

$$R \geq \frac{2g(h_A - h_B)}{g}$$

$$R \geq \frac{2g(25 - 15)}{g}$$

$$R \geq 20\text{m}$$

$$R_{\min} = 20\text{m}$$

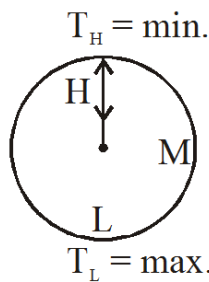
38. **Ans (4)**

Critical speed at A, B and C are

$$v_A = \sqrt{5Rg}; v_B = \sqrt{3Rg}; v_C = \sqrt{Rg}$$

$$v_A : v_B : v_C :: \sqrt{5} : \sqrt{3} : \sqrt{1}$$

39. Ans (1)



$$\frac{T_L}{T_H} = 4 \quad T_L = 4T_H$$

$$T_L - T_H = 6mg$$

$$3T_H = 6mg$$

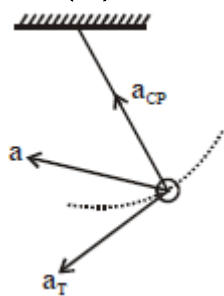
$$T_H = 2mg$$

$$\text{also } T_H + mg = \frac{mv_H^2}{R}$$

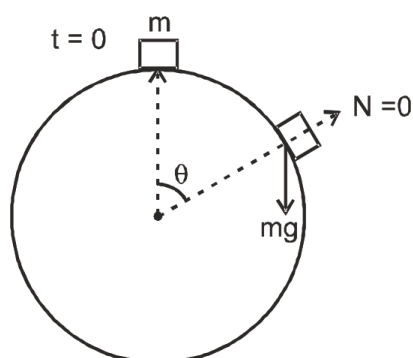
$$2mg + mg = \frac{mv_H^2}{R}$$

$$V_H = \sqrt{3Rg} = \sqrt{3 \times \frac{10}{3} \times 10} = 10 \text{ m/s}$$

40. Ans (3)



41. Ans (2)



at loose contact $N = 0$

$$mg \cos \theta = \frac{mv^2}{R} \quad \frac{mv^2}{R} \quad \frac{mv^2}{R} \quad \frac{mv^2}{R} \quad \dots (1)$$

from energy conservation

$$mgR(1 - \cos \theta) = \frac{1}{2} \frac{1}{2} \frac{1}{2} \frac{1}{2} mv^2 \quad \dots (2)$$

from (1) & (2)

$$\cos \theta = \frac{2}{3} \Rightarrow \sin \theta = \frac{\sqrt{5}}{3}$$

tangential acceleration

$$g \sin \theta = \frac{\sqrt{5}g}{3}$$

42. Ans (3)

By Newton laws

$$mg - N = \frac{mv^2}{R}$$

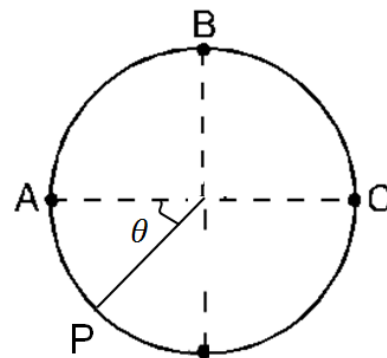
$$mg - \frac{mv^2}{R} = N \quad \text{where } m_1 g_1 v = \text{const.}$$

$$R_A < R_B$$

$$\left(mg - \frac{mv^2}{R_A} \right) < \left(mg - \frac{mv^2}{R_B} \right)$$

$$N_A < N_B$$

43. Ans (2)



Let Particles meet at P after t time

$$\theta = \omega t, \quad \omega = \frac{v}{r}$$

$$\text{For A } \theta = \frac{1}{r} t \quad \dots (1)$$

$$\text{For B } \theta + \frac{\pi}{2} = \frac{2.5}{r} t \quad \dots (2)$$

$$\text{For C } \theta + \pi = \frac{2t}{r} \quad \dots (3)$$

$$(2) \div (3)$$

$$\theta = -3\pi$$

$$\frac{\theta_B}{\theta_C} = \frac{-3\pi + \frac{\pi}{2}}{-3\pi + \pi} = \frac{5}{4}$$

44. Ans (2)

$$F_C = mk^2 r t^2$$

$$a_C = k^2 r t^2 = \frac{v^2}{r} \Rightarrow v = krt$$

$$a_t = \frac{dv}{dt} = kr$$

$$Ft = mkr \Rightarrow P = \vec{F} \cdot \vec{v} \quad (\because \vec{F}_C \cdot \vec{v} = 0)$$

$$P = \vec{F}_t \cdot \vec{v} = mkr \times krt$$

$$= mk^2 r^2 t$$

45. Ans (2)

In uniform circular motion resultant horizontal force on the car must be towards the centre of circular path.

